

The Tour d'Horizon of Data Law Implications of Digital Twins

Industry Considerations

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One may be familiar with Marvel Comics character “Iron Man,” who designed his suit using a 3D hologram model [1]. This fictional idea of a virtual replica has now culminated into a reality in the form of ‘Digital Twins.’ Bridging the gap between the physical and virtual realms, ‘Digital Twins’ has become a buzzword in numerous industries from healthcare to aerospace to energy to manufacturing.

A Digital Twin (“DT”) is a virtual replica of real-world applications. Cars, humans, supply chain, telecom networks, business processes, and even entire cities can have their DTs. Global DT market size is projected to grow from USD 17.73 billion in 2024 to USD 259.32 billion by 2032, forecasting a CAGR of 39.8% [2].

DTs are data intensive applications and may have several implications under various data laws globally. This article aims to provide a broad overview of the data law considerations (privacy and sectoral laws) for digital twins across various jurisdictions.

1 UNDERSTANDING AND CATEGORIZING DTs

The first known use of a DT was by the National Aeronautics and Space Administration in the 1960s for the Apollo 13 mission, where 15 simulators were used to replicate the physical spacecraft's conditions [3]. The Digital Twin Consortium¹ has defined a DT as “*an integrated data-driven virtual representation of real-world entities and processes, with synchronized interaction at a specified frequency and fidelity.*”² The international standard ISO/IEC 30173:2023 on *Digital twin – Concepts and terminology* also provides for a definition of digital twins.³ However, the acceptance of this definition is subject to the acceptance of the international standard by the respective countries.

To put simply, a common man may perceive a DT as a “replica” of a physical product or process, continuously updated with real-time data from the physical counterpart. Typically, depending upon the utility and type of the DT, it may be facilitated or supported by various technologies including Internet of Things (“IoT”) devices like sensors for data collection, cloud computing for data storage and retrieval, Artificial Intelligence (“AI”) for data analytics and immersive technologies (augmented, virtual and mixed reality) to create virtual replicas [4].

Against this backdrop, it is pertinent to note that DTs may be broadly classified as follows.

¹ Official website of the Digital Twin Consortium. <https://www.digitaltwinconsortium.org>

² Definition of a Digital Twin, Digital Twin Consortium.
<https://www.digitaltwinconsortium.org/initiatives/the-definition-of-a-digital-twin/>

³ ISO/IEC 30173:2023 Digital twin – Concepts and terminology.
<https://www.iso.org/standard/81442.html>

1.1 PRODUCT DT

A Product DT spans the entire lifecycle of a physical product, from design to disposal, using bidirectional data flows. Industry research distinguishes between a “component twin” (a digital replica of a single part) and an ‘asset twin’ (a model combining two or more component twins to analyse interactions) [5]. For example, a car engine DT would be a component twin, while a DT of the entire car—engine, battery, wheels, ignition, and cooling systems—would be an asset twin. A few use case examples of a Product DT are discussed below.

1.1.1 DEFENSE AND AVIATION

In the defense and aviation sector, DTs are used for designing, testing, and monitoring *products* like fighter jets, commercial aircraft, and drones. The use of Product DT in this sector can enhance longevity by prompting timely repairs and maintenance, as well as improve durability through testing and adjustments. Here are a couple of use cases.

Intelligent Engines

Rolls Royce's *IntelligentEngine* vision uses digital technology to design, test, and maintain aircraft engines in the ‘digital sphere.’ The company employs a DT of the physical engine to simulate potential test scenarios.⁴ Additionally, their *Engine Health Management* feature transfers data from physical engines to operational centers for ongoing performance monitoring.⁵

DT Army

The Australian Army and Royal Australian Air Force contracted McKinsey & Co. to create a DT of the army and two air bases for training and decision-making on personnel deployment.⁶ While details remain unclear due to security concerns, DTs in defense may be used to (i) test fighter jets under various simulated weather conditions, (ii) monitor asset performance, and (iii) simulate and test defense systems like missiles [6].

1.1.2 HEALTHCARE

DTs of specific organs are utilized to monitor the functioning of the respective organ and to simulate its behavior under varying conditions. These DTs may be classified as Product DTs, as their scope is confined to individual organs and their respective functions. Overall, DT technology has seen growing adoption in healthcare, with companies like Siemens and Microsoft offering DT-based services [7].

⁴ Intelligent Engine – Our vision for the future of aircraft power, Rolls Royce – Our Stories, 2020.
<https://www.rolls-royce.com/media/our-stories/discover/2020/intelligentengine-explainer.aspx>

⁵ Intelligent Engine Health Monitoring, Rolls Royce- Our Stories, Jan 2019.
<https://www.rolls-royce.com/media/our-stories/discover/2019/intelligent-engine-health-monitoring.aspx>

⁶ Chirgwin R (2022), Digital twins for Defense a \$16m win for McKinsey, itnews.
<https://www.itnews.com.au/news/digital-twins-for-defence-a-16m-win-for-mckinsey-578380>

Living
Heart and
Living
Brain

Dassault Systèmes created DTs for two organs – the brain and the heart – aimed at early diagnosis and treatment of related ailments.⁷ The Living Brain initiative uses patient-specific data, such as brain structure, biomarkers (e.g. cortex thinning, white matter), and electrophysiological data, to diagnose conditions like Alzheimer's and epilepsy. The Living Heart initiative models a patient's heart, replicating the electrical system that controls heart muscle contraction.

1.2 PROCESS DT

Process DTs replicate entire systems and the relationships between process steps, using data from interconnected physical components. As per industry research, “process twin” (Process DT), creates a digital environment with multiple interconnected components, assets, and system twins. For instance, Process DT for the entire shop floor can give a comprehensive view of all machines, enabling more informed decisions about repairs without disrupting operations.⁸ A few use-cases are discussed below.

1.2.1 MANUFACTURING

A Process DT of a manufacturing unit may be created to streamline and manage the process. A few examples are below.

Factory of
the Future

Mars Inc. used DTs to address package over-spilling by feeding sensor data from physical machines into analytics models which allowed real-time monitoring and adjustments. Later, Mars partnered with Accenture to optimize global production through (i) real-time machine monitoring and (ii) performance prediction using analytics [8].

Azure
Digital
Twins

Microsoft offers Azure Digital Twins as a platform-as-a-service for creating DTs of physical environments. The DT presents a 3D visual model of the environment, with an IoT hub device enabling live data transfer from sensors.⁹

⁷ Meet Your Virtual Twin, Dassault Systèmes.

<https://www.3ds.com/virtual-twin/life-sciences-healthcare/meet-your-virtual-twin>

⁸ The Process Digital Twin: A step towards operational excellence, Microsoft Corporation – Digital Twin Vision.

<https://info.microsoft.com/rs/157-GQE-382/images/Digital%20Twin%20Vision.pdf>

⁹ What is Azure Digital Twins?, Microsoft.

<https://learn.microsoft.com/en-us/azure/digital-twins/overview>

1.2.2 HEALTHCARE

Precision medicine

The DT generated is based on personalized data, including age, gender, weight, height, genetic profile, organ functions (renal, liver), blood analysis, and personal habits (e.g. smoking). Predictions on drug efficacy and reactions are made based on the DT. For ExactCure's DTs, both historical data on the drug's impact and the patient's personal data are collected to make accurate predictions on medication effects.¹⁰

1.3 NETWORK DT

A Network DT replicates a network comprising of connections among multiple nodes, such as those in smart cities, telecommunications, logistics, railroads, supply-chains, oil fields, electric grids, water resources management, etc. This twin is continuously updated with real-time data from the network and sensors attached at various nodes. Network DTs enhance operational efficiency by allowing network operators to test proposed changes on the virtual network before applying them to the real one [9]. In addition, they can help anticipate impact of any type of disruption, often related to natural calamities or breakdown of certain nodes, on the entire network and identify alternate paths for the fastest recovery.

1.3.1 TELECOMMUNICATIONS AND ENERGY

Telecommunications

Network DTs for telecommunications replicate wireless links, fibers, sites, and their generated data, updated in real-time to enable proactive problem detection. For example, Ceragon uses data analytics to examine how different components interaction and simplifies the collected data through a data interface.¹¹ Similarly, Ericsson employs Building Information Modeling ("BIM") to create Network DTs, integrating existing component DTs, business logic, project-specific data, and the DT of the installed base, sites, and networks to create a 3D model for testing various "what-if" scenarios.¹²

Energy

A Network DT may also be used for the purposes of replicating an electric grid. Such Network DTs are used for testing emergency grid operations management to combat the issue of wildfires [10]. The data related to the transmission lines, the terrain and the transfer of electricity is taken into consideration for creating such DTs.

¹⁰ Digital Twins, Exactcure Dassault Systèmes.
<https://3dexperienancelab.3ds.com/en/projects/life/exactcure/>

¹¹ Ceragon's Network Digital Twin Solution, Ceragon.
<https://www.ceragon.com/solutions/ceragon-network-digital-twin>

¹² Kirac, E. Using digital twins to be in control of your network assets, Ericsson.
<https://www.ericsson.com/en/blog/2022/5/using-digital-twins-to-be-in-control-of-your-network-assets>.

1.3.2 SUSTAINABILITY PROJECTS

Water Management	DTs of water bodies, like rivers, can simulate changes in water temperature, levels, and flow rates, aiding in water management and flood risk prediction. For example, the Lushan Water Supply Company in China created a DT of its water systems, using it to simulate optimizations before implementation, which helped minimize costs and operational disruptions [11].
Lights Out	A World Economic Forum white paper highlights how DT technologies have improved both productivity and environmental impact. For example, Western Digital in Malaysia used over 1,000 IoT sensors on 500 pieces of equipment and 15 utility systems to optimize energy and water use, cutting greenhouse gas emissions and waste. By implementing a 'lights out' process with automated systems and robotic transport, operations could be remotely controlled and shut off when not needed, significantly reducing machine idle time and saving energy [12].

1.4 COMMUNITY DT

Community DTs are virtual models of subdivisions, boroughs, or even entire urban areas, created with drones and updated in real-time. They capture every detail, from buildings to alleyways, facilitating efficient urban planning. Two prominent use cases are given below.

1.4.1 SMART CITIES

Orlando DT	The Orlando Economic Partnership created a DT mapping 800 sq. miles of the Orlando region, with 40 square miles mapped with high detail. The DT layers' multiple datasets covering infrastructure, and traffic patterns, and includes mapping the flow of new talent in the workforce. ¹³ This provides insight into the economic development of the Orlando region and enables informed decision making.
Virtual Singapore	Singapore is the first country to have a DT, created by scanning the terrain with aircraft and automobiles using cameras and lasers. ¹⁴ Data for the DT was taken from public agencies, legacy sources, and real-time inputs. ¹⁵ The DT helps monitor the city

¹³ Orlando Regional Digital Twin, Orlando Economic Partnership.

<https://business.orlando.org/l/orlando-regional-digital-twin/>

¹⁴ Walker, A. Singapore's digital twin – from science fiction to hi-tech reality, Infrastructure Global.

<https://infra.global/singapores-digital-twin-from-science-fiction-to-hi-tech-reality/>

¹⁵ Customer Stories - Virtual Singapore, Dassault Systèmes.

<https://www.3ds.com/insights/customer-stories/virtual-singapore>

efficiently, improve building sustainability, and map potential solar projects.¹⁶ Additionally, Virtual Singapore would also be categorized as a Community DT (explained above) providing insights into day-to-day operations.

2 DATA FLOWS

DTs are data intensive and process large volumes of non-personal and personal data which is transmitted from the physical asset. A common concern across all applications is the heavy reliance of DT on input and real time data. All factors of performance of the DT including efficacy, capacity, replications and implementation are directly co-related to the quality and quantity of data of the original application which is made available to the DT (“**Input Data**”) which flows from the physical assets which the DT replicates. Given the significance of such Input Data, this section delineates the key stages of data flow, with a particular emphasis on data acquisition, recognizing its pivotal role in the analysis set forth in this article.

2.1 ACQUISITION

In most DT models, the data flows directly from the physical asset on a real time basis or is procured from the physical asset. Depending on the type of the DT application, appropriate quality Input Data is identified. For example, where a replica of a human organ is the DT application, the Input Data is likely to be personal data (identification of the individual, medical records, physiological data, genetic data). Such data may be procured either directly from the individual or through a healthcare provider. On the other hand, in case of engine DT, the Input Data may be procured directly through the IoT sensors attached to the physical engine. One of the primary considerations for the smooth functioning of the DT model is to ensure that the Input Data is procured from an identified source in accordance with applicable laws governing data collection, sharing and processing.

2.2 VALIDATION, STRUCTURING AND ANONYMIZATION

Accuracy and quality of data is vital, especially when sourced from multiple origins. Without proper validation and filtration of procured data; erroneous or biased data may negatively impact the simulations and predictions of the DT. Subsequent to collection, the data should be validated and verified; and classified and categorized in alignment with the purpose of use. The collected data may also be anonymized based on the requirement of the operations sought to be conducted. There may be different processes and mechanisms adopted for this process depending upon the DT application and its utility.

¹⁶ Digital Twinning a Country, Singapore, World Geospatial Industry Council.
<https://wgicouncil.org/digital-twinning-a-country-singapore/>

2.3 DEVELOPMENT AND DEPLOYMENT

Once the DT is deployed and the application is in use, there is ongoing processing of the data which was fed into the DT. Depending upon the type of application, there may also be real-time data collection and processing. Such data may include feedback and user inputs, data from the operations environment, etc.

3 CONSIDERATIONS UNDER DATA LAWS

Regulatory framework governing data privacy and protection across various jurisdictions can place restrictions on the scope of data collection and processing. Some of the key legal nuances are discussed below.

3.1 STAKEHOLDERS

Privacy laws across jurisdictions typically apply to three categories of stakeholders.

3.1.1 DATA CONTROLLER

Individuals or entities which determine the “why” and “how” of the collection, processing, storage and sharing of the data. Across most jurisdictions, the Data Controller is responsible for the highest level of legal compliance. In DT applications, depending on DT model and functionality, the DT developer and the entity which is deploying the applications could be considered Data Controllers.

3.1.2 DATA PROCESSOR

The entity which acts on behalf of the Data Controller to perform data processing tasks according to the controller’s instructions. Data Processor does not have autonomy over the purpose or method of processing but simply executes tasks assigned by the Data Controller. Outsourcing or backend service providers, network/cloud platforms where DT is hosted, DT operators, etc. could be considered Data Processors.

Therefore, depending on the type of DT application, the role of each contributing stakeholder in the development and deployment of the application, the party would be designated as the Data Controller or Data Processor.

3.1.3 DATA SUBJECT

In most jurisdictions, Data Subject refers to a natural person and not a juristic person or an object. In the context of personal data (i.e. any information that relates to or can identify an individual), the individual to whom the data concerns is the Data Subject. The Data Controller and Processors are responsible to the Data Subject for personal data collected and processed. Personal data of Data Subjects would generally be processed in the creation of human or organ DTs. In such use cases, Data Subjects are granted various rights under multiple jurisdictions, including the right to withdraw consent for the processing of their data. Upon withdrawal of consent, the Data

Controller may be obligated to delete such personal data, ensuring compliance with applicable data protection regulations. With reference to the Use Cases mentioned above, the use of personal information in Healthcare related DTs, data protection and privacy regulations would get triggered given that the Input data would presumably contain personal data of the individual concerned. However, if the data is related solely to an aircraft engine, or a network, and does not include data related to an individual, then such data protection and privacy regulations may not be applicable. These would be governed under specific legislations, if any, related to the use of non-personal data.

Where non-personal data or technical data is collected and processed, such data may not relate to any specific natural person. There may not be a data subject in these cases. This may include use-cases where the DT is that of a car engine or of an electronic grid, where no personal data related to an individual would be required to create the DT.

3.2 DATA AGREEMENTS

At certain stages of the DT data lifecycle, data may flow between relevant stakeholders. For instance, (i) during the procurement stage, data is acquired from a source, (ii) during the processing stage, the data may be shared with a third-party Data Processor and (iii) lastly, post-deployment, the final data may be shared with relevant stakeholders. In such arrangements, two distinct data agreements often govern the transfer of data:

- *Data Controller and Data Subject Agreement:* When personal data is collected from an individual (Data Subject), the Data Controller—potentially the DT provider—is required to provide notice and obtain explicit consent as per most jurisdictions. The notice must clearly outline the purpose for data collection and processing, as well as inform the Data Subject of their rights (e.g. access, rectification, and erasure).
- *Data Controller and Data Processor Agreement:* When the Data Controller engages a third-party Data Processor to process personal data, the relationship must be formalized through a contract. This contract should specify the security standards to be upheld, the responsibilities and liabilities of each party in case of a breach, and the mechanisms for audits and oversight to ensure compliance.

Such agreements are crucial for delineating roles, determining the ownership of data, security obligations, maintaining transparency, and adhering to legal requirements throughout the DT data lifecycle. For instance, agreements between a Data Controller and a Data Processor typically include clauses to ensure the secure handling of data and prevention of unauthorized access: (i) confidentiality of data and its employees, (ii) enabling access to data to authorized individuals within each party's organization based on their roles and responsibilities, (iii) technical security measures such as multi-factor authentication, encryption etc., (iv) compliance with specific data security standards and periodic compliance audits.

The creation of a DT is a data intensive task. As visible from the examples mentioned above, there are many companies who operate at a B2B level and create DTs for other businesses. Here, data agreements as well as confidentiality agreements become ever more relevant to protect crucial operation related data of businesses opting for DT related services from a third-party.

3.3 CROSS BORDER DATA TRANSFERS AND LOCALIZATION

In most DT applications, the data flows and transfers may not be limited to one jurisdiction. For instance, the procurement of Input Data may be from a different jurisdiction; or the data processing may be in a different jurisdiction than where the DT is deployed. This may be particularly relevant where a company that intends to create a Process DT of its manufacturing units, which may spread across multiple jurisdictions. Let us take an illustrative example - a company, headquartered in California, may have a manufacturing unit in both India and China, and may want to combine the data of both these units to gain insights for operational efficiency. The Process DT (including the DT of the China and India units) may be created and accessed by the company in California. In such cases, the laws governing the data import/export and processing in all the relevant jurisdictions would be applicable.

From this perspective, it is significant to note that certain jurisdictions have adopted cross-border data transfer restrictions or data localization requirements. For instance, transferring personal data outside the European Union (“EU”) involves stringent compliance requirements. One approach is through an adequacy decision, where the European Commission recognizes that a non-EU country ensures an adequate level of data protection and permits the transfer of personal data. In the absence of such a decision, data transfers may be based on contractual arrangements.¹⁷ Likewise, transfer of certain sectoral data from India (such as payments data, insurance data, geospatial data, financial services data, etc.) is either prohibited entirely or a copy of such sectoral is required to be maintained in India. In addition to this, jurisdictions may not permit the transfer of data related to defense, in the interest of national security (discussed below under Data Localization).

3.4 DATA MINIMIZATION, PROCESSING AND RETENTION LIMITATIONS

One of the core principles across data protection laws is that only minimum and necessary data should be collected/processed, and used for specific purposes and for a duration justified for fulfilling such purpose or for which permissions have been obtained. Therefore, in DT applications, it must be ensured that only a minimum amount of data is procured/processed and no unused dataset is being hosted on DT systems. This is especially integral in applications that are collecting and processing personal data, where the Data Subject has to consent to each of the specified purposes of processing and retention of their personal data. Further, the Data Controller may be obligated to erase the personal data of the Data Subject, if the Data Subject

¹⁷ This includes the “Standard Contractual Clauses” and “Binding Corporate Rules.”

has the right to withdraw their consent and has exercised this right. However, personal data may be retained after the withdrawal of consent if the law mandates the retention of that specific category of data for a specified period of time.

4 LEGAL IMPLICATIONS AND RISK MANAGEMENT

The legal risks and implications under data laws would depend on the type of data and the jurisdiction. Firstly, the legal risks and compliances associated with personal data is typically higher than non-personal data. Furthermore, if data used in the DT application is “*sensitive personal data*” under local data laws, the DT application may be subject to higher compliances. Secondly, it must be noted that data laws are country-specific, hence, DT application should be assessed on a jurisdictional basis. Key legal nuances under data laws to keep in mind are detailed below.

4.1 VIOLATION OF PRIVACY

If personal information of an individual is collected to create a DT, privacy laws are triggered. Further, certain jurisdictions such as Australia¹⁸, EU¹⁹, Saudi Arabia²⁰ further categorize certain types of personal data as sensitive personal data (such as health data, genetic data, etc.), in which case, higher compliances would be triggered. Under the currently applicable Indian data protection law (“**Current Indian Data Law**”), only sensitive personal data or information is subject to regulatory compliance, while other categories of personal data remain outside its scope.²¹ However, India is transitioning from the Current Indian Data Law to a new data protection framework (“**Proposed Indian Data Law**”).²² The Proposed Indian Data Law eliminates the distinction between sensitive personal data and personal data, applying its provisions uniformly to all personal data, which is broadly defined as any data that identifies an individual or relates to an identifiable individual.²³

¹⁸ “sensitive information”, Section 6, Privacy Act, 1988, Australia.

¹⁹ Article 9, Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation) (“EU-GDPR”).

²⁰ Article 1(11), Personal Data Protection Law (Royal Decree No. (M/19) dated 09/02/1443 AH corresponding to 16/09/2021), Kingdom of Saudi Arabia (“PDPL, Saudi Arabia”).

²¹ Information Technology Act, 2000 (“IT Act, India”) and the Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Data or Information) Rules, 2011, India (“IT Rules, India”) (collectively “Current Indian Data Law”).

²² Digital Personal Data Protection Act, 2023, India (“DPDPA, India”) and the Draft Digital Personal Data Protection Rules, 2025 (“DPDP Rules, India”) (collectively “Proposed Indian Data Law”). [Please note that the DPDPA, India has not been enforced as of the date of writing this article. The Indian government has issued the DPDP Rules, India for public consultation. Once the rules are finalized, DPDPA, India and DPDP Rules, India would come into enforce and would replace the Current Indian Data Law.]

²³ Section 2(t), DPDPA, India.

4.1.1 POTENTIAL RISKS

In the event, the Data Controllers in the DT ecosystem fail to undertake appropriate compliances under privacy laws such as – (i) seeking consent of Data Subjects, (ii) ensuring Data Subject rights, or (iii) protecting the personal data, the privacy laws may be violated and penalties may be enforced. For example, personal data collected to create an organ DT, as specified in a notice to the Data Subject, cannot be repurposed without explicit consent.

Sharing this personal data with a third-party should also be based on consent obtained from the Data Subject. Additionally, DT providers acting as Data Controllers must ensure personal data is securely stored and accessible only to authorized personnel, particularly for sensitive data related to health, genetics, and biometrics. Data Subjects may also bring an action against the DT developer/deployer for compensation.

This risk is heightened especially where sensitive personal data is involved. In some jurisdictions, only the Data Controller is held liable under the law, even if the Data Processor is at fault. For instance, under the Proposed Indian Data Law, the Data Fiduciary (akin to a Data Controller) is responsible for compliances under the data law, regardless of whether the processing of the personal data is undertaken by a contracted Data Processor.²⁴ To mitigate this risk, DT developers/deployers may include an indemnity clause to address breaches or data leaks caused by the Data Processor. This safeguards the DT developers/deployers interests in case a monetary penalty is imposed on them for such incidents.

There is an increasing global concern over subjecting Data Subjects to automated decisions. This may be relevant in the context of Manufacturing DTs or Healthcare DTs, where workers or patients (as the case may be) may be subject to decisions made by AI algorithms. To counter this, it must be ensured that (i) the Data Subjects are explicitly informed about the process of the decision making, and (ii) critical decisions impacting the Data Subject are subject to human review and oversight. For instance, a doctor or medical professional should not place absolute reliance on the diagnosis given by an organ DT and should critically review the diagnosis themselves.

4.1.2 IMPACT

If appropriate compliances under personal data protection laws are not undertaken, Data Controllers in the DT application may be subject to statutory penalties. In most jurisdictions,

²⁴ Section 8(1) read with Section 8(5), DPDPA, India.

including India²⁵, EU,²⁶ Singapore,²⁷ heavy monetary penalties are triggered for non-compliance. For instance, in the EU, administrative fines can reach up to EUR 10 million or (for undertakings) 2% of the total worldwide annual turnover of the preceding financial year, whichever is higher.²⁸ Similarly, under Proposed Indian Data Law penalties vary based on the contravention, ranging from INR 10,000 (approx. USD 118) to INR 250,00,00,000 (approx. USD 29.59 million).²⁹ However, if the personal data is irreversibly anonymized, then the privacy law compliance and penalties would not be triggered.

Additionally, specifically for Manufacturing DTs, the domestic laws may require the DT provider to comply with technical standards such as the “*Automated Systems and Integration – Digital Twin Framework for Manufacturing*.”³⁰ These standards establish a comprehensive framework that, among other provisions, outline the creation of functional entities for data assurance, security support, and access control.³¹ Furthermore, they provide guidelines for defining information attributes within a DT³² and for the functionality of networks and information exchange processes across the DT system.³³

4.2 DATA SUBJECT RIGHTS

Privacy laws in various jurisdictions grant Data Subjects rights vis-à-vis their personal data including, amongst others: the right to access, right to be forgotten, right to portability and right

²⁵ Information Technology Act, 2000 (“IT Act, India”) and the Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Data or Information) Rules, 2011, India (“IT Rules, India”); Digital Personal Data Protection Act, 2023, India (“DPDPA, India”). [Please note that the current data protection framework in India is regulated by the IT Act read with the IT Rules. The government has enacted the DPDPA, but has not issued the rules under it and hence, the DPDPA is yet to be enforced. Once the DPDPA is enforced, it would replace the IT Rules.]

²⁶ Chapter 8, EU-GDPR.

²⁷ Section 48J, Personal Data Protection Act, 2012, Singapore.

²⁸ Chapter 83(4), EU-GDPR.

²⁹ Schedule I, DPDPA.

³⁰ ISO 23247-1:2021 Automated Systems and Integration – Digital Twin Framework for Manufacturing - Part 1: Overview and General Principles.
<https://www.iso.org/standard/75066.html>

³¹ ISO 23247-2:2021 Automated Systems and Integration – Digital Twin Framework for Manufacturing - Part 2: Reference Architecture.
<https://www.iso.org/standard/78743.html>

³² ISO 23247-3:2021 Automated Systems and Integration – Digital Twin Framework for Manufacturing - Part 3: Digital representation of manufacturing elements.
<https://www.iso.org/standard/78744.html>

³³ ISO 23247-4:2021 Automated Systems and Integration – Digital Twin Framework for Manufacturing - Part 4: Information Exchange.
<https://www.iso.org/standard/78745.html>

to correct and update. These rights may be exercised against the Data Controller and the Data Processor (in some jurisdictions).

4.2.1 POTENTIAL RISKS

The quantity of data that the Data Collector or Data Processor may handle in Healthcare Use Cases are high. Here, considering the high volume of personal data (such as historical data) and sensitivity of the data (such as genetic data), it would be difficult for the Data Collector or Data Processor (as the case may be) to enable a Data Subject to exercise the above-mentioned rights due to operational difficulties. Furthermore, individual rights pose different challenges. For instance, jurisdictions such as EU³⁴ and Brazil³⁵ provide the Data Subject with the right to portability. In the EU, this right allows Data Subjects to receive their personal data from a Data Controller in a machine-readable format and transmit it to another Data Controller. Additionally, they can request direct transmission between Data Controllers, where feasible. However, jurisdictions such as Canada and India, do not provide for such portability rights to Data Subjects. In a cross-jurisdictional context, this may act as a hindrance for the Data Subject to exercise their rights. As mentioned above, historical data from multiple Data Subjects may be used to create the DT and accessing and porting this data may be a challenge for the Data Controller.

Similarly, the Proposed Indian Data Law grants a unique right to Data Subject - the right to nominate another individual who may exercise the Data Subjects Rights in the event of death or incapacity. From an operational perspective, Data Controllers are burdened with ascertaining the validity of nomination.

4.2.2 IMPACT

The high volume and sensitivity of healthcare data may make it challenging for Data Controllers and Processors to enable Data Subject rights, especially in cross-jurisdictional contexts. While jurisdictions like the EU and Brazil provide portability rights, others, such as Canada and India, do not, which may create operational and compliance hurdles for DT operations involving multiple jurisdictions.

Further, it is the obligation of the Data Controller to ensure that the Data Subject is not deprived of control over their personal data, especially in the context of Healthcare DTs. Here, the collection of personal data should be based on explicit consent of the Data Subject and there should be clear mechanisms in place to ensure that the Data Subject can exercise the rights granted under the relevant jurisdictions. Even if there is a difference in the rights granted to the Data Subject (such as the example of data portability), it should at minimum be ensured that common rights such as the right to access and correct the personal data are made available to the Data Subject.

³⁴ Article 20, EU-GDPR.

³⁵ Article 18, Lei Geral de Proteção de Dados, Federal Law no. 13,709/2018, Brazil ("LGPD, Brazil").

4.3 DATA LOCALIZATION

In some jurisdictions such as India, Singapore and Russia, certain types of data (personal data or otherwise) are prohibited from being transferred or even accessed outside the specific jurisdiction. For instance, Russia³⁶ imposes restrictions on even access to certain categories of data. In 2024, the USA released an executive order that prevented access to bulk sensitive personal data and government related data by countries of concern³⁷. In India, local geospatial laws prohibit access to certain sensitive geospatial data known as “negative attributes.” This prohibited category of geospatial data includes defense areas. Additionally, foreign entities are explicitly prohibited from accessing geospatial data which is finer than the prescribed thresholds.³⁸ Hence, DTs in the defense sector may have restrictions to ensure that the data is stored within the territorial limit of the country concerned.

4.3.1 POTENTIAL RISKS

Defense related DTs may be created by an external private company, which may have multiple offices across the globe. Data related to defense is sensitive and may include geospatial data and data on weapon capabilities, military sites and even military supply chain. If such data is not permitted to be transferred outside the relevant jurisdiction, it would limit the accessibility to such data by the DT. In which case, the DT may not be able produce the desirable or precise results.

4.3.2 IMPACT

Data localization requirements for certain categories, such as defense, may restrict cross-border data transfers, hindering real-time data updates essential for DT applications. Further, DT applications impacted by data localization laws would have to be deployed at domestic level, i.e. the servers and data centers hosting the applications should be located in the same region as where the Input Data is collected and the DT application is deployed. Hence, this would involve significant infrastructure costs to the developer and the deployer of DT to ensure that the Input Data subject to localization is not being transferred outside the specific country.

4.4 DATA TRANSFERS

Data laws in most jurisdictions permit the cross-border transfer of personal data subject to certain conditions, which include that the recipient country offers adequate or same level of

³⁶ Federal Law No. 149-Fz of July 27, 2006 On Information, Informational Technologies and the Protection of Information, Russia.

³⁷ Executive Order on Preventing Access to Americans’ Bulk Sensitive Personal Data and United States Government-Related Data by Countries of Concern issued by the White House, dated February 28, 2024.

³⁸ Guidelines for acquiring and producing Geospatial Data and Geospatial Data Services including Maps, 2021, India; Office Memorandum on Clarifications/Compliance points in pursuance to Geospatial Guidelines dated November 28, 2022, India.

protection (as the transferor country) to the personal data and/or explicit consent has been obtained from the Data Subject for the transfer of data.

4.4.1 POTENTIAL RISKS

Under data laws, safety and transfer of health related information has been given specific attention. For instance, a legislation specific to the state of Ontario (Canada) permits the transfer of personal health information only if at least one of the seven conditions prescribed are met. These conditions, amongst others, include specific consent of the individual or if it is necessary for the provision of healthcare to the individual, provided that the individual has not objected to such disclosure.³⁹ Interestingly, Saudi Arabia has a specific exception that permits the transfer of personal data in cases of extreme necessity such as to preserve the life of the Data Subject, or to prevent, examine, or treat a disease.⁴⁰ This may permit the transfer of data for the creation of human or organ DT (as may be required).

In some countries, the sectoral data laws also prescribe data transfer restrictions. For certain Product DTs such as cars, aircrafts (in the Aviation Use Case), geospatial data may also be collected and may be subject to sectoral laws related to geospatial data. For instance, in India, as previously stated, the export of specific categories of geospatial data is restricted,⁴¹ and China has a strict licensing regime to permit entities to survey and map the territory.⁴²

On similar lines, data related to defense may be subject to confidentiality requirements due to its critical link to national security. For instance, in the US, the Secretary of Defense carries out a “Defense Information Assurance Program.” The objective of this program includes ensuring the authenticity and confidentiality of the defense information infrastructure. Its strategy encompasses conducting vulnerability and threat assessments, organizing exercises, and simulating scenarios to counteract information warfare threats effectively.⁴³

Similarly, in China, there is a specific legislation to guard “state secrets.” As per this law, state secrets encompass information related to national defense and activities of the armed forces.⁴⁴ It prohibits the unauthorized carrying, transmission, posting, or transport of documents or

³⁹ Section 50, Personal Health Information Protection Act, 2004, Ontario, Canada.

⁴⁰ Article 29(3), PDPL, Saudi Arabia.

⁴¹ Guidelines for acquiring and producing Geospatial Data and Geospatial Data Services including Maps, 2021, India; Office Memorandum on Clarifications/Compliance points in pursuance to Geospatial Guidelines dated November 28, 2022, India.

⁴² Surveying and Mapping Law of the People's Republic of China, 1992, China [Our analysis is based on the English translation of the law].

⁴³ §2224, Chapter 131, Part IV, Subtitle A – General Military Law, Title 10- Armed Forces, US.

⁴⁴ Article 8, Law of the People's Republic of China on Guarding State Secrets (1988).

materials designated as state secrets beyond the country's borders without approval from competent authorities.⁴⁵

In addition to this, specific restrictions imposed on the transfer of personal data in the relevant jurisdictions must also be taken into account. For instance, the Proposed Indian Data Law empowers the government to restrict the transfer of certain specified categories of personal data to a foreign state or an entity or person controlled by such foreign state.⁴⁶ Further, identified "significant" data fiduciaries (akin to Data Controllers) may be required to process identified categories of personal data within the territory of India.⁴⁷ These additional restrictions, once implemented, would need to be carefully considered to ensure compliance and seamless operational continuity.

4.4.2 IMPACT

Given the large datasets used in DT applications, the DT developers/deployers will need to identify the compliances and transfer restrictions for each type of data within the Input Datasets. Additionally, data transfer agreements with Data Processors must be executed to ensure alignment with these compliance obligations. This is a compliance burden. From an implementation perspective, even if there are higher restrictions for transfers of certain datasets and not for others, the entire DT application would be affected.

4.5 DATA STORAGE & SECURITY

DT utilizes cloud computing technology to integrate data between the physical products and the virtual replica. Hence, we understand that data related to the DT may be stored in a cloud. The cloud computing technology may be owned, operated and managed by the DT provider or may be outsourced to a third-party. Here, most DT providers offer cloud computing solutions as a part of the DT. However, there may be situations, where storage is outsourced. In addition to this, there is a requirement to notify a designated authority in the event of a data breach/leak. In some jurisdictions such as EU⁴⁸ and India⁴⁹, the Data Subject should also be notified. Further, some jurisdictions such as India also have cyber security incident reporting requirements even if no personal data is affected.

4.5.1 POTENTIAL RISKS

Cloud storage solutions may be susceptible to cyber-attacks resulting in leaks and/or breaches of data. All Use Cases would be at a risk of such a threat. However, the sensitivity of the data lost

⁴⁵ Article 26, Law of the People's Republic of China on Guarding State Secrets (1988).

⁴⁶ Rule 14, DPDP Rules, India.

⁴⁷ Rule 12(4), DPDP Rules, India.

⁴⁸ Article 34, EU-GDPR.

⁴⁹ Section 8(6), DPDP, India.

would vary. For instance, health and defense data are categorized as sensitive in comparison to data about a machine's health (such as in a Manufacturing DT). Further, it may not be guaranteed that the cloud service provider and the Data Processor may ensure security of the data hosted on cloud servers.

4.5.2 IMPACT

A data breach or data leak of the cloud may happen, which could impact the privacy of an individual (under Healthcare) or loss of proprietary information (under other Use Cases).

Cloud storage solutions may be deployed by the DT provider directly or may be procured from a third-party through a contractual arrangement. In both the cases, the liability for any breach/leak would be on the DT provider, being the Data Controller. Therefore, it becomes relevant for Data Controllers to contractually ensure that cloud service providers and Data Processors (who have access to the data stored on cloud) maintain adequate security, adopt safeguards and monitor mechanisms and report data breaches.

4.6 DATA FIDELITY

As mentioned above, DTs are data intensive and the accuracy of data used for the creation of these DTs is of critical importance. Inaccuracy in the Input Data or historical data may lead to inaccurate or erroneous outcomes.

4.6.1 POTENTIAL RISKS

In the Healthcare Use Case, a DT may be used for diagnosis and treatment. If such a DT is fed with inaccurate, incomplete or inconsistent records of patient data, it may result in faulty and incorrect diagnosis and treatment of the disease or predictions related to medicines to be administered. Inaccurate outcomes resulting from inaccurate, incomplete or inconsistent Input Data or historical data may cause harm to human health (in case of Healthcare Use Case) or may impede operational efficiency in other Use Cases.

AI forms an intrinsic part of the DT model and may be influenced by biases from the past. In the context of Healthcare, data may be subject to such bias which may impact the outcome. For instance, a study reported by MIT revealed that AI takes a "*short-cut*" by basing the diagnosis on learnt demographic categorizations.⁵⁰ Furthermore, research indicates that discrimination in healthcare can persist through historical records of diagnoses made by individual doctors decades ago.

These biases, embedded in the data from 20 years ago, can inadvertently influence modern AI systems. Even if the discriminatory factors are no longer prevalent in society today, their

⁵⁰ Trafton, A. Study reveals why AI models that analyze medical images can be biased, MIT News <https://news.mit.edu/2024/study-reveals-why-ai-analyzed-medical-images-can-be-biased-0628>.

remnants in historical data may manifest in AI-driven decisions, perpetuating inequities.⁵¹ In addition, there may be scenarios where doctors do not act independently—such as prescribing specific medications due to external influences like pharmaceutical guidelines or institutional policies—yet these prescriptions are recorded as data for training AI systems. Such data, reflecting constrained or non-autonomous decision-making, can inadvertently introduce biases into AI models, potentially leading to skewed or suboptimal recommendations.

From a privacy context, it becomes relevant to ensure that the fidelity of data is safeguarded from unauthorized access or modifications at all stages, which may compromise the integrity of the data used in the DT.

Network DTs of energy grids are used to prevent forest fires, and if such a DT is subject to inaccurate data, it may result in inaccurate analysis. Such analysis could be counterproductive and may not prevent forest fires. Similarly, in Manufacturing, an inaccurate outcome may harm the operational efficiency of the manufacturing process and in remote situations, the output may be defective which may have a ripple effect on consumers and human life.

4.6.2 IMPACT

Data protection and privacy laws impose an obligation on the Data Controller (and Data Processors in some cases) to ensure that the personal information of the Data Subject collected is accurate. For instance, Australia mandates that the personal information collected is “*accurate, up-to-date and complete*,” and since the Australian law does not distinguish between a Data Processor and a Data Controller, this obligation would be applicable to any entity that collects and/or processes personal data of a Data Subject.⁵²

On the other hand, in Saudi Arabia, the Data Controller has to ensure the accuracy of the data before it is processed.⁵³ The Data Controller is mandated to notify the Data Subject if the personal data is corrected, updated or completed⁵⁴ and in addition to this, there are compliance regulations that a Data Controller is mandated to comply with while correcting the data.⁵⁵ The above-mentioned obligations would be applicable in the Use Cases under Healthcare. The liability of the DT provider in each jurisdiction would depend on whether it is classified as a Data Controller or Data Processor for the DT, as well as the specific liabilities imposed on these roles by the jurisdiction.

⁵¹ Carey, S., Pang, A., & Kamps, M. (2024) Fairness in AI for healthcare, Future Healthcare Journal (Vol. 11(3)), pp 2.

<https://doi.org/10.1016/j.fhj.2024.100177>

⁵² Principle 10, Schedule 1, Privacy Act 1988, Australia.

⁵³ Article 14, PDPL, Saudi Arabia.

⁵⁴ Article 17, PDPL, Saudi Arabia.

⁵⁵ Article 22, Implementing Regulation of the Personal Data Protection Law, Saudi Arabia.

For other Use Cases, it is likely that there may be contractual obligations with indemnity clauses to mitigate losses caused due to inaccurate outcomes from the DT.

5 CONCLUSION AND TAKE AWAYS

To conclude, we propose a few risk mitigation measures to ensure the smooth, efficient and secure operation of DT technology in the Use Cases mentioned above.

5.1 HUMAN OVERSIGHT

The decisions provided by a DT are automated. Hence, there should be vigilant human oversight before such decisions are implemented on the physical product. This becomes especially relevant in the Use Case of Healthcare where automated decisions related to the diagnosis of an ailment may be implemented on a patient. Furthermore, the DT provider should have a designated ‘Data Protection Officer,’ as required in multiple jurisdictions, to ensure that any complaints from Data Subjects are addressed promptly and effectively.

5.2 EXPERTS AS A RESOURCE

The creation of a DT should not be an entirely automated process. It is suggested that inputs from experts in the required field are taken for the creation of the concerned DT. A hybrid approach should be adopted to combine human knowledge and machine learning to create DTs. An expert’s insight would be valuable to curate the “*correct process recipe*” for the DT. For instance, the construction of energy efficient building needs the aid of human knowledge and experience in extreme terrains. Experts can accurately determine the data that might be required for such a project. Experts act as a check for the actions proposed to be taken to create a DT [13].

5.3 DATA CLASSIFICATION

Depending on the type of DT being created, an analysis should be undertaken to understand the category of datasets involved such as non-personal data, personal data and sensitive personal data. Identification and classification of data would aid in data compliance by various stakeholders involved.

5.4 ENTITY CLASSIFICATION

To further mitigate risks and to ensure compliance with applicable laws, all stakeholders—such as the DT provider, service provider, storage provider, back-end service provider, and consumer-facing entities (e.g. hospitals offering precision medicine services in collaboration with a DT provider)—must assess their roles as Data Controllers or Data Processors under the relevant jurisdiction. This assessment should account for the nature of the data involved, for instance; whether it pertains to human health (sensitive personal data) or machine health (non-personal data). Proper identification of roles ensures that each stakeholder fulfills its legal obligations under applicable laws, thereby strengthening compliance frameworks and reducing liability risks.

5.5 MAPPING

The flow of data in a proposed DT should be mapped before the process of creation is started. Accordingly, a jurisdiction-focused analysis, depending on the type of DT being used, should be undertaken to apprise oneself of the laws applicable to such operations. This would ensure informed decision making with respect to prospective operations. Additionally, businesses should implement technological, operational, and organizational measures to ensure compliance with applicable laws. Conducting an impact assessment taking into consideration the nature of operations, types of personal data involved, and the potential risks of harm is essential for identifying vulnerabilities and mitigating legal and regulatory risks.

5.6 DATA QUALITY AND ASSURANCE

Data is the core source for the creation of DTs. Therefore, steps should be taken to ensure that the source of data collection is verified at both levels – for Input Data as well as historical data. Further, regular data audits – both internal and external should be conducted to ensure robust compliance with relevant data protection laws. Further, to ensure the security of data, regular data protection impact assessments may also be conducted.

5.7 DATA AGREEMENTS

Data Agreements at both levels between the (i) Data Controller and the Data Subject, and (ii) the Data Controller and its Processors are essential. These determine the contractual relationship between the parties, duties and responsibilities and rights and liabilities.

5.8 HONORING RIGHTS

As noted above, honoring Data Subjects' rights can be challenging due to the high volume and sensitivity of personal data in Healthcare DTs. To address this, a data interface, such as a dashboard, could be provided by the Data Controller, allowing Data Subjects to manage their personal data directly. This approach would streamline operations and enhance transparency.

5.9 SECURITY AND INSURANCE

Robust cybersecurity measures must be implemented by the DT provider, including encryption, access controls, and continuous monitoring to protect personal data. A comprehensive cyber response plan should also be in place to swiftly address any security breaches, alongside clear breach reporting protocols to ensure compliance with relevant regulations. Additionally, securing appropriate cyber insurance can provide financial protection against potential data breaches or cyberattacks, helping to manage the associated risks effectively. Overall, the synchrony required for the efficient functioning of a DT should also be reflected in the functioning of the DT provider's operations as well.

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